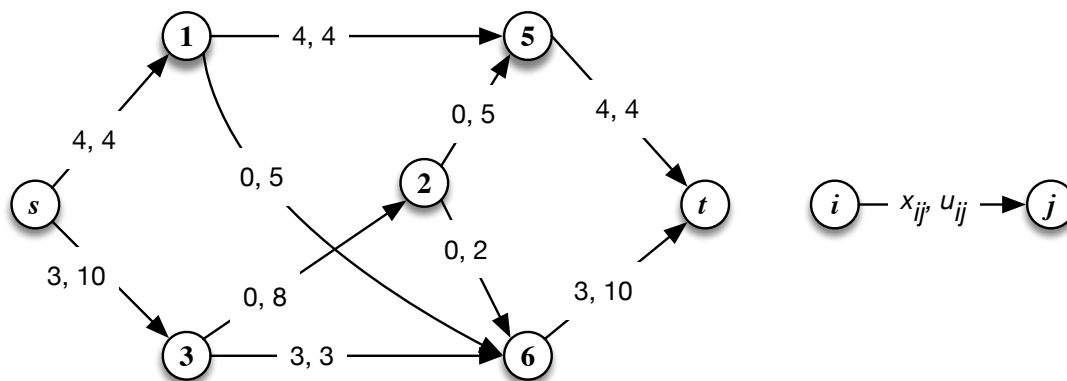


First part : Closed book

Total score: 21 pts. Minimum score to pass: 11 pts. Tentative scores: 1) 4 pts, 2) 5 pts, 3) 5 pts, 4) 3 pts.

1. Determine the maximum flow in the following graph starting from the given flow (x). Report the intermediate steps and the final minimum capacity cut. What is the worst case complexity of the used algorithm?

Does the maximum flow changes if the capacity of arc (1,5) increases? Why?



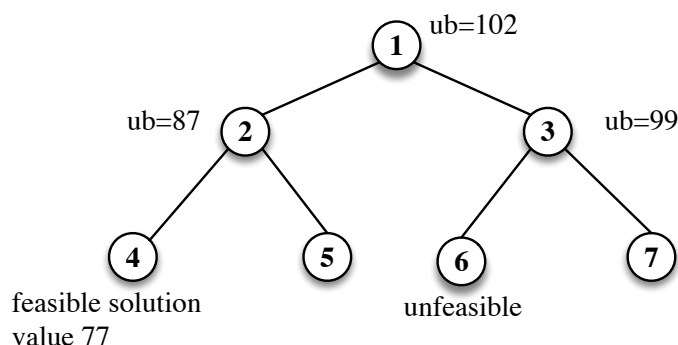
2. Consider the following linear programming problem:

$$\begin{aligned} \max \quad & -2x_1 + x_2 \\ & -6x_1 - 2x_2 \leq -3 \\ & x_2 \leq 4 \\ & 2x_1 + 3x_2 \leq 15 \end{aligned}$$

Write the dual and the complementary slackness equations.

For each of the solutions the solutions $x' = \begin{bmatrix} -3/2 \\ 6 \end{bmatrix}$ and $x'' = \begin{bmatrix} 3/2 \\ 4 \end{bmatrix}$ derive the corresponding complementary dual solutions and discuss their primal and dual feasibility arguing their optimality.

3. Consider the an Integer Linear Programming problem of the form $\max\{cx : Ax \leq b, x \in \mathbb{Z}_+^n\}$. During the computation of the optimal solution via Branch and Bound, the enumeration tree looks as follows:



Nodes 5 and 7 are “open”. What are the possible value for the upper bound in node 5? What are the outcomes of the computation of the upper bound in node 7 that close the overall computation?

4. Consider the vectors $a^1, a^2, \dots, a^r \in \mathbb{R}^n$. Write the definition of the cone finitely generated by $a^1, a^2, \dots, a^r$. Is it a convex set? Justify your answer.

AMPL questions: 4 points.

in the rear of this paper

Write the statements in the AMPL modeling language for the following definitions:

- a set I
- a set J
- a positive integer variable x_{ij} with $i \in I$ and $j \in J$.
- a parameter b_j with $j \in J$.

Then using the defined variables and parameters write in the AMPL modeling language the following constraint:

$$\sum_{i \in I} x_{ij} \leq 5$$

$$\forall j \in J : \text{if } b_j = 15$$

ANSWER IN THE SPACE BELOW